THE SOURCE OF A CLOSED IMMERSION INTO A COUNTABLY INDEXED AFFINE FORMAL ALGEBRAIC SPACE IS OF THE SAME TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own wording 'countably index' for 'countably indexed'; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Countably indexed affine formal algebraic spaces as formal spectra of complete weakly admissible rings. Same chapter, Lemma lemma-countably-indexed, invoked at proof lines 6543-6545 and 6549-6551. Identifies both spaces with formal spectra and fixes the kind of ring involved.
- A closed immersion is representable and affine, and the type ascends along a representable affine morphism. Same chapter, Lemma lemma-property-goes-up-affine-morphism, invoked at proof lines 6545-6549. Gives that the source is affine and countably indexed.
- Closed immersions into a McQuillan affine formal algebraic space correspond to taut maps with closed kernel and dense image. Same chapter, Lemma lemma-closed-immersion-into-McQuillan, invoked at proof lines 6551-6554. Gives tautness and dense image of the ring map.
- A continuous taut map with dense image between these rings is open and surjective. Same chapter, Lemma lemma-dense-image-surjective, invoked at proof lines 6554-6557. Concludes that the ring map is a quotient map.

Source reference. Original lemma 0ANQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 6138–6138. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2493–2493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0APT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1455–1455. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4316–4316. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 0ANA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4953–4953. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be a countably indexed affine formal algebraic space over S. Let $f : Y \to X$ be a closed immersion of formal algebraic spaces over S. Then Y is a countably indexed affine formal algebraic space and f corresponds to $A \to A/K$ where A is an object of $WAdm^{count}$ (Section 0ANA) and $K \subset A$ is a closed ideal.*

Proof. By Lemma 0AN1 we see that $X = \mathrm{Spf}(A)$ where A is an object of $WAdm^{count}$. Since a closed immersion is representable and affine, we conclude by Lemma 0AKP that Y is an affine formal algebraic space and countably index. Thus applying Lemma 0AN1 again we see that $Y = \mathrm{Spf}(B)$ with B an object of $WAdm^{count}$. By Lemma 0ANQ we conclude that f is given by a morphism $A \to B$ of $WAdm^{count}$ which is taut and has dense image. To finish the proof we apply Lemma 0APT. □